Level 1 Data Collection & Exploration Answers

By Liz Sneddon

Exercise 1:

Ākonga will have their own answers, but here are examples.

1) I'm interested in co-curricular activities at school because I've heard that being active helps with mental health.
2) I'm interested in gathering data on cellphones as I want to see which models and apps are popular and commonly used to help with my business studies course.

Exercise 2:

1) Elastic type: Categorical, Nominal
   Length: Numeric, Continuous
   Number of marbles: Numeric, Discrete
   Stretched length: Numeric, Continuous
   Weight: Numeric, Continuous
   Obstacles: Categorical, Nominal
2) Minutes: Numeric, Continuous
   Gender: Categorical, Nominal
   AgeGroup: Categorical, Ordinal
   Stridelength: Numeric, Continuous

Exercise 3:

1) Poll, because ākonga were asked for their opinion in one question.
2) Survey, because ākonga were asked multiple questions in a questionnaire.
3) Experimental study, because ākonga were randomly allocated to two groups. One group did the exam on paper and the other on a computer, hence a treatment was applied.
4) Observational study, because data was collected after observing the number of shots that went into the hoop.

Exercise 4:

1) B. Every person in the United States.
2) D. A selection of 3,294 people who bought the album.
3) B. All of the website visitors.
4) C. A selection of voters of different ages.

Exercise 5:

1) Survey. The respondents are ākonga.
2) Observational study. The participants are the water samples.

Exercise 6:

Ākonga are aged between 14 – 16 years old, taking Level 1 NCEA Maths, a mixture of ethnicities and cultures, different hair lengths, eye colours, etc.

Exercise 7:

1) Biased, Self-selected
2) Random, Stratified
3) Random, simple random
4) Biased, Convenience

Exercise 8:

1) Primary, because we collect data ourselves by measuring heights and ages.
2) Secondary, because Statistics NZ has collected the data, not me.
3) Secondary, because the website collected the data, not me.

Exercise 9:

1)

Time to run 100 meters	Stopwatch
Weight of an apple	Kitchen scales
Capacity of liquid in a bottle	Measuring jug or cylinder
Volume of music on the radio	Decibel meter
Angle of a wheelchair ramp	Clinometer or app on phone

2) Different scales might be slightly different in the weights that they provide, as they may not be calibrated exactly the same, or be different brands with different mechanisms etc. This leads to the data being inconsistent.

3) So that the lengths are accurately measured (the true length is measured), and they measurements are consistent where all pens have their lids on.

Exercise 10:

Sample size	189
Location of data collection	Baystate Medical Centre, Springfield, Mass.
Year data was collected	1986
Purpose of the study	Identify risk factors associated with giving birth to low birthweight baby (weighing less than 2500 grams).

Variables	**Description**
LowBirthWeight	No (birthweight $<$ 2500), Yes (birthweight $\geq$ 2500)
MothersAge	Age of mother in years
Race	Race of mother (white, black, other)
MotherSmoke	Smoking during pregnancy (Yes or No)
FTV	Number of physician visits during first trimester
BirthWeight	Birth weight in grams

Exercise 11:

1)
- Some ākonga may be injured or feel uncomfortable doing physical tests.
- Data shouldn’t be shared with anyone else, and when data is recorded, don’t record the names of ākonga.
- Care and respect should be shown by not recording names of ākonga or making inappropriate comments (such as ākonga not having name branded shoes e.g. Nike).

2)
- The wording of questions needs to be respectful and non-judgmental (e.g. not being negative about people who are vegans).
- Ask questions that allow ākonga to feel comfortable asking about any cultural or religious protocols that they follow.

- Names of ākonga will need to be recorded (e.g. to know who has allergies so that they don’t get served incorrect food), but this information needs to be kept private and confidential apart from those who need to know (e.g. the person/people who are preparing or serving the food).
- Get consent from respondents to share the information about allergies etc. with appropriate people.

Exercise 12:

1) Shorter time, less reliable
2) Longer time, more reliable

Class Exercise:

- Every helicopter will be slightly different. For example, when cutting out the helicopter, ākonga won’t cut exactly on the lines, and if there is more surface area, this likely makes the weight a little higher, increasing the speed that the helicopter drops, reducing the time that it takes to drop. To manage this, we could use one single helicopter for all of the different trials.
- Each time the helicopter is dropped with differing length rotor blades, the results will be different due to variation. To help manage this variation, we could take several repeated measurements for each length of rotor blades. Then we can find the mean of these measurements, which will be a more accurate measurement of the time it takes to drop.
- Having the same person using a stopwatch is important, as each person has a slightly different reaction time (difference between the time the helicopter touches the ground and the time it takes to press stop on the stopwatch).
- The helicopter should be dropped by the same person from the same location and position, so that any differences in the distance that the helicopter drops is as similar as possible for each trial. If the distance that the helicopter drops is longer, then the time it takes to fall is likely to be slightly longer. This will mean that the data is more consistent.
- Factors such as the wind can affect the time for the helicopter to fall. Wind can give the helicopter a little more lift, increasing the time, or it might push the helicopter down which would decrease the time. To manage this, it would be better if we did the tests inside the Gym rather than outside, as we can close the doors of the gym meaning that wind won’t have a significant effect.

Exercise 13:

1) It is important to measure the ingredients accurately as changing the amount of ingredients could change the consistency, or outcome of the brownies, making them chewy or crispy if mismeasured.
2) The ingredients need to be added in the order given in the recipe because if one ingredient is added out of order the outcome of the brownies could be wrong. The brownies might be too dry, or the ingredients may not incorporate properly.
3) This instruction was added because the quantity can change if the recipe is doubled or halved, and not everyone has the same-sized bowls. If the ingredients did not fit well, when mixed some ingredients could be lost.

4) Another source of variation is the temperature of the oven the brownies are baked, if the temperature is too low or high it could affect the brownies, if the temperature was lower than the brownies are likely to be undercooked, and if the temperature is higher, it is likely that they will burn.
Another source of variation is if the dry ingredients are sifted, this could affect the consistency of the brownies.

5)
- Have a countdown for ākonga to start the race, 3-2-1 so that every ākonga starts running at the same time as the person on the stopwatch pressing start.
- Have one person recording the time to run 100m, so that the data is more accurate. Each person has a different reaction times (the time between someone saying the number 1 and the person pressing start on the stopwatch), so if the same person is measuring all trials, the data will be more consistent.
- Have all ākonga in the sample wearing the same type of clothing and shoes. If they are all wearing a school PE t-shirt and school shorts, then the weight of the clothing will be consistent. For example, if someone was wearing a school jumper as well as their PE gear, as they have more weight being carried by their clothes, the time it would take them to run 100m is likely to be slightly longer than if they weren't wearing a jumper.

6)
- Every apple on a tree is different, due to the amount of sunlight (one side of a tree gets more sunlight than other sides), the amount of rain, nutrients, etc. However, it is not possible to manage this as when we buy a bag of apples, they have been packaged by someone else who doesn't take this into account.
- The same scales should be used to measure the weights of all the apples, and the scales should be zeroed before the measuring begins. This means that the variation between different scales will be controlled and managed. This means that the data on the weight of apples will be more consistent.
- Different varieties of apples are likely to have different weights. For example, granny smith apples tend to be larger than some other varieties. To manage this, we will make sure that the bag of apples that we buy all have the same variety of apple. This will help the weights of the apples to be more consistent in their measurements.

7)
- The device that is playing the radio channels needs to be the same and have the same setup/settings used. For example, if the volume of a radio station's song is measured from different speakers, then this would lead to more variation in the volumes measured, which would be due to the speakers rather than the radio stations.
- The volume of each radio station that is measured, should all be measured when a song is being played, not when adverts are being played or when radio hosts are talking, as there are likely to be different due to who is speaking/singing rather than a valid comparison between stations. Advertisements usually tend to be a little louder than songs as this is one of the marketing techniques used to get the attention of listeners to the radio station, and advertising is one of the main sources of income for radio stations.
- The same decibel meter should be used to measure the volume, and we plan to measure the volume in one single time period, so that the meter isn't being turned on and off between each sample, as this may lead to differences in decibel levels due to the equipment rather than the radio station volumes. The volume data will be more consistent.

Exercise 14:

1) Instructions will be different for each classroom and ākonga, but should include things such as:
   - Stepping around desks and chairs (this may include accurate measurements)
   - Turning left and right
   - Moving forward or backwards
2) Reflections from ākonga on how to improve their instructions.

Exercise 15:

1)
   - The variable gender is not an accurate representation of the population. It only allows for people to select male/female, and it is not clear whether this represents the gender people identify as, or whether it represents the sex that people are born with (I suspect it sex is the most likely for athletes running a marathon). To better manage this in the future, organisers should be more specific about whether they are collecting information on sex or gender, and if it is gender, then more categories should be included (such as LGBTQ+).
   - The weather during the race may affect the time and stride length for the athletes. For example, if it rains during the race, this is likely to increase the time it takes to complete a marathon as the ground will get a little more slippery and muddy. However, it is not possible to control the weather, but the people collecting the data could try to collect data from a marathon where the weather was dry and sunny throughout the whole race, as this would lead to data that is more consistent.
   - I am not sure how the average stride length was calculated. Athletes will have longer stride length on flat ground and shorter stride length on hills as the gradient of the ground affects the speed that athletes run. To manage this possible variation, it is best if an athlete's average stride length is calculated on the same flat piece of ground for all athletes, so that the data is more consistent.
2)
   - From the metadata information provided, it appears that the temperatures were all collected in the same location at Auckland Airport. This is good as it means that the temperatures are more consistent, as the temperatures close to the ocean or further inland are likely to differ.
   - As the data on temperatures are collected every month (actually it is likely to be collected daily), it is likely that the same equipment is used, which would lead to more consistent measurements.

Exercise 16:

Elastic type	Length (cm)	Number of marbles	Stretched length (cm)	Weight (grams)	Obstacles
Wide	69	6	89	24.8	Yes
Wide	69	8	92	46.4	No
	69	10	95	58	Yes
Cord		12	98	69.6	No
Cord		14	99	81.2	No
Cord		16	102	92.8	Yes
Cord	69	18	106	104.4	Yes
string	69	20	108	116	
Narrow	68.03	10	116	58	Yes

Exercise 17:

Ākonga will have their own answers.

Exercise 18:

1)	**Data display**	Bar Graph
	Number of variables	One
	Variable definition(s)	Types of fruit
	Data types	Categorical
2)	**Data display**	Dot Plot
	Number of variables	Two
	Variable definition(s)	Weight (kg), Species of Kiwi (GS, NiBr, ToK)
	Data types	Numerical (weight), Categorical (species)
3)	**Data display**	Scatter Graph
	Number of variables	Two
	Variable definition(s)	Engine Size, Price ($)
	Data types	Numerical
4)	**Data display**	Frequency Table
	Number of variables	Two
	Variable definition(s)	Age groups, Flavour of ice cream
	Data types	Categorical
5)	**Data display**	Bar Graph
	Number of variables	One
	Variable definition(s)	Weight (kg)
	Data types	Numerical
6)	**Data display**	Strip graph
	Number of variables	One
	Variable definition(s)	Species
	Data types	Categorical
7)	**Data display**	Pie chart
	Number of variables	One
	Variable definition(s)	Favourite type of movie
	Data types	Categorical
8)	**Data display**	Time Series
	Number of variables	Two
	Variable definition(s)	Sales (thousands), Quarters
	Data types	Numerical
9)	**Data display**	Box Plot
	Number of variables	One
	Variable definition(s)	Carat (weight)
	Data types	Numerical
10)	**Data display**	Dot Plot
	Number of variables	One
	Variable definition(s)	Price ($)
	Data types	Numerical
11)	**Data display**	Frequency Table
	Number of variables	Two
	Variable definition(s)	Number of days with rain, Number of weeks
	Data types	Numerical

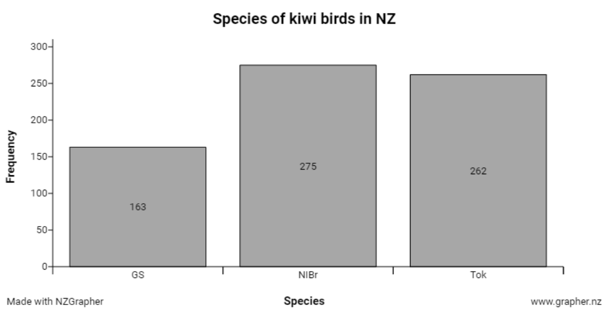
12)	**Data display**	Stem and Leaf plot
	Number of variables	Two
	Variable definition(s)	Before and After, speed (km/hr)
	Data types	Categorical, Numerical
13)	**Variable(s)**	Gender
	Data types	Categorical
14)	**Variable(s)**	Age groups
	Data types	Categorical Ordinal
15)	**Variable(s)**	Minutes
	Data types	Numerical Discrete
16)	**Variable(s)**	Stride Length
	Data types	Numerical Continuous
17)	**Variable(s)**	Minutes, Age group
	Data types	Numerical Continuous, Categorical Ordinal
18)	**Variable(s)**	Gender, Age group
	Data types	Categorical Nominal / Ordinal

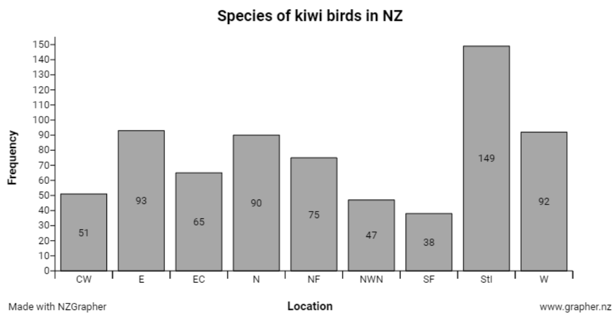
Exercise 19:

1) Ākonga will select the dataset
2) Ākonga will fill out the tables themselves.

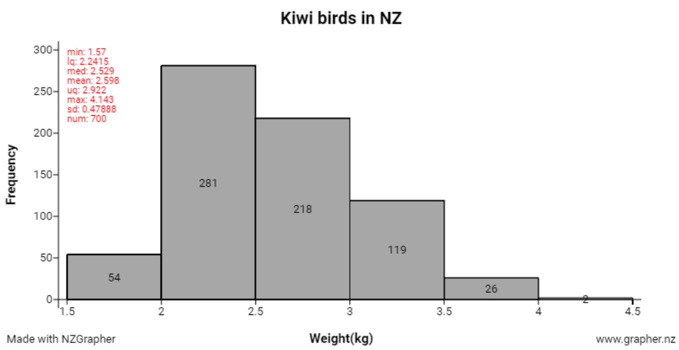
Data point	Species	Gender	Weight	Height	Location
20	GS	F	3.719	46.8	NWN
40	Tok	M	2.457	36.5	NF

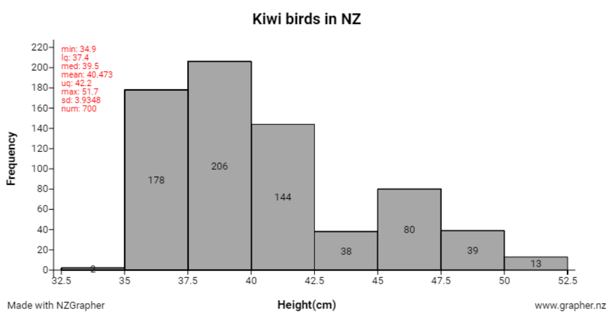
3)





4)





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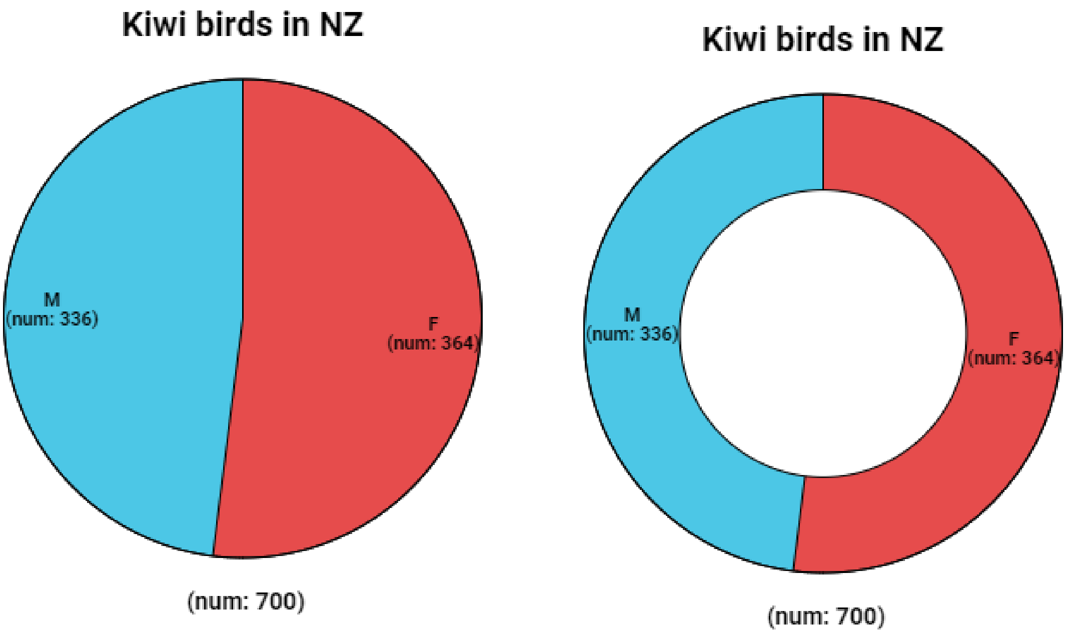

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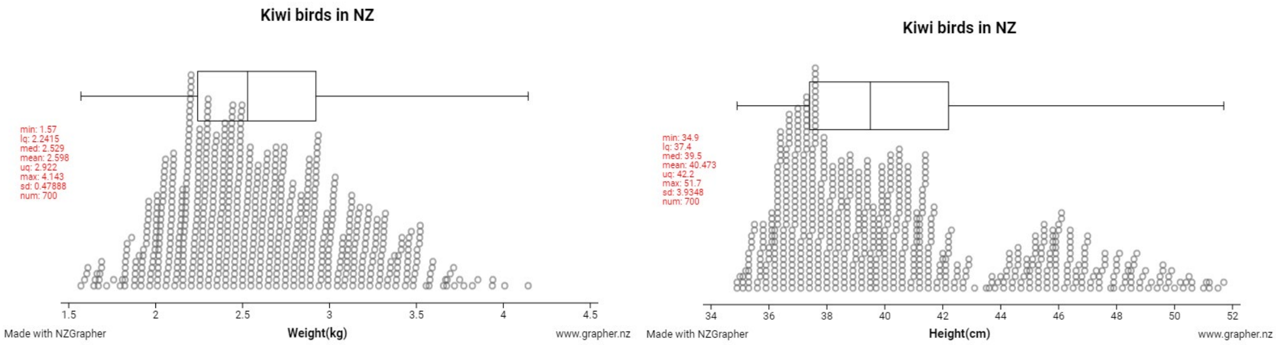
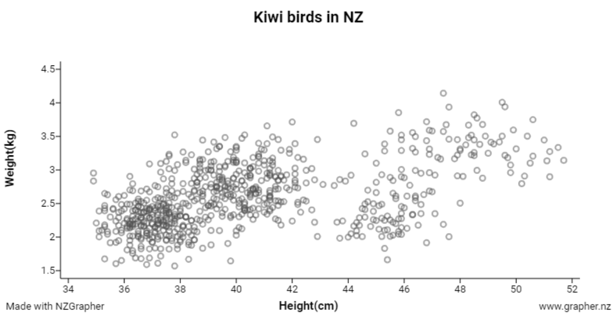
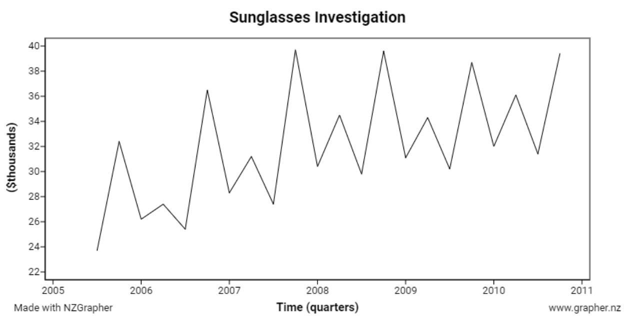


7)



8)



Exercise 20:

Ākonga will have a variety of graphs from their exploration.

Exercise 21:

1)	**Data types**	Height – Numeric, continuous Country – Categorical, nominal
	Data display(s)	Stem and leaf plot, dot plot, box and whisker graph
	Investigation type	Comparison
2)	**Data types**	Height – Numeric, continuous Weights – Numeric, continuous
	Data display(s)	Scatter graph
	Investigation type	Relationship
3)	**Data types**	Probability of winning – Numeric, continuous
	Data display(s)	Frequency table, relative frequency table, bar chart, long-run-relative frequency graph
	Investigation type	Experimental probability
4)	**Data types**	Time – Numeric, discrete Number of people – Numeric, continuous
	Data display(s)	Time Series graph
	Investigation type	Time Series
5)	**Data types**	Number of hours studying – Numeric, discrete Number of credits – Numeric, discrete
	Data display(s)	Scatter graph
	Investigation type	Relationship
6)	**Data types**	Sports (play, not play) – Categorical, nominal Spped – Numeric, continuous
	Data display(s)	Stem and leaf plot, dot plot, box and whisker graph
	Investigation type	Comparison

Exercise 22:

Suggestive words	Definitive words
Maybe	*Prove*
Suggest	*Definitely*
Likely	*Will*
May	*Must*
Might	*Have to*
Can	*etc*
Tends	
Possible	
Etc	